

APPLICATIONS
**EFFECT OF THE DURATION OF
BREASTFEEDING ON THE LIFETIME OF
CHILDREN IN LESOTHO**

D.J. de Waal, Zeleke B. Worku[†], P.C.N. Groenewald.

University of the Free State, P.O. Box 339, Bloemfontein, 9300, South Africa.

[†]Technikon Natal, Eastend, Durban.

Key words: Bayes estimates; Farlie-Morgenstern bivariate model; survival function.

Summary: A bivariate Farlie-Morgenstern type model is constructed to determine the relationship between duration of breastfeeding and mortality rate of children under the age of five years in Lesotho. The data are from a survey done during October 1994 on 4001 live births in the Maseru district within five years of the survey. The official five year mortality rate for Lesotho is 13.5% and it is suspected that a short duration of breastfeeding could be a leading indicator for a high mortality rate. Avoiding a causal interpretation, we were able to determine that there is a positive relationship between duration of breastfeeding and mortality rate. For instance, a child who received 9 months of breastfeeding has a probability of 0.961 of reaching the age of five compared to a probability of 0.884 for a child who received one month of breastfeeding, conditional on both surviving beyond the period of breastfeeding.

1. Introduction

A survey (Worku, 1995) was conducted during October 1994 in the Maseru district of Lesotho in an attempt to determine the leading causes of the high mortality rate among infants in Lesotho. According to Lesotho Department of Health figures (Lesotho Bureau of Statistics, 1994), 13.5% of children alive at

birth die before the age of five. Out of 55200 children under the age of five (born from September 1989 to September 1994), a random sample of size 4001 was taken. Of these children, 71% were rural while the remaining 29% were from the urban parts of the Maseru district. The Maseru district was stratified into several zones for data collecting using proportional allocation with respect to population size. Procedures of systematic random sampling were used to identify eligible households. At each household the mother was interviewed in Sesotho.

Among a large number of possible indicators of mortality rate (such as age of mother, income, literacy and postnatal care) recorded during the interview, the duration of breastfeeding was considered an important one. To determine the effect of the duration of breastfeeding on the probability of survival of under five year old children, a model has been constructed to describe the joint distribution of the two variables X (duration of breastfeeding in years) and Y (lifetime of child in years).

There are a number of reasons why most of the standard regression models in survival analysis are not appropriate in this case. Of the 4001 observations, 253 are censored in both X and Y while 3355 are censored in Y alone. Of the remaining uncensored observations we have 211 where $X = Y$ and 182 where $X < Y$. Normally the restriction that $X \leq Y$ could have been eliminated by a suitable transformation, but in this case there is a positive probability mass on the line $X = Y$. This requires a mixed joint distribution. Furthermore, the sample was drawn from children under the age of five years of which nearly 84% were censored. Most of these observations censored in Y will eventually reach a value well beyond five, but since our population of interest is the under

five year olds, we must construct a lifetime distribution between zero and five, and will then put a positive probability mass on $Y = 5$, representing all children surviving to the age of five years and beyond.

In section 2 we fit the marginal distributions and construct a joint distribution based on the Farlie-Morgenstern family with parameter of association α . The Bayes estimators of all the parameters are obtained in section 3 by means of the Gibbs sampler. In section 4 the survival curve, conditional on duration of breastfeeding, is derived.

2. Construction of the model

To avoid the complication of one variable's sample space depending on the other variable, we will consider the two variables X (duration of breastfeeding), and $Z = Y - X$, the survival time after termination of breastfeeding. The aim now is to construct a joint distribution of X and Z with specified marginals.

2.1 Marginal distribution of X

The marginal distribution of X is assumed to be Weibull with parameters θ and β . In Grummer-Strawn (1993) an adjusted Weibull model is proposed. The adjustment is due to the fact that not all children are initially breastfed and therefore the survival rate at time zero is not 100%. However, for the Lesotho data, every child was breastfed, so no adjustment was necessary. For some other proposed models see, for example, McDonald (1981) and Vanderhoeft (1982).

The likelihood function is given by

$$L(\theta, \beta | \underline{x}) = \prod_{i=1}^n [f(x_i)]^{\delta_i^x} [1 - F(x_i)]^{1 - \delta_i^x} \quad (2.1)$$

where

$$f(x) = \frac{\beta}{\theta^\beta} x^{\beta-1} e^{-(x/\theta)^\beta}, \quad x > 0, \quad (2.2)$$

$$F(x) = 1 - e^{-(x/\theta)^\beta}, \quad x > 0. \quad (2.3)$$

$$\text{and } \delta_i^x = \begin{cases} 1 & \text{if breastfeeding has stopped} \\ 0 & \text{if breastfeeding is still continuing.} \end{cases}$$

The maximum likelihood estimators of θ and β are $\hat{\theta} = 0.4324$ and $\hat{\beta} = 2.3123$. The estimated mean and variance are 0.383 years (4.60 months) and 0.031 years (4.45 months) respectively. In figure 1 the Kaplan-Meier estimate (Kaplan and Meier (1958)) of the cumulative distribution function (cdf) is shown with the cdf of the fitted Weibull distribution. Notice that the duration of breastfeeding was only recorded to the nearest month.

2.2 The marginal distribution of Z

The density function of Z is continuous between zero and five, but with positive probability masses at $Z = 0$ and at $Z = 5$, representing, respectively, the proportion of children dying before normal termination of breastfeeding and the proportion of children surviving for at least five years beyond termination of breastfeeding. It is assumed that Z is exponentially distributed, but truncated at 5 years.

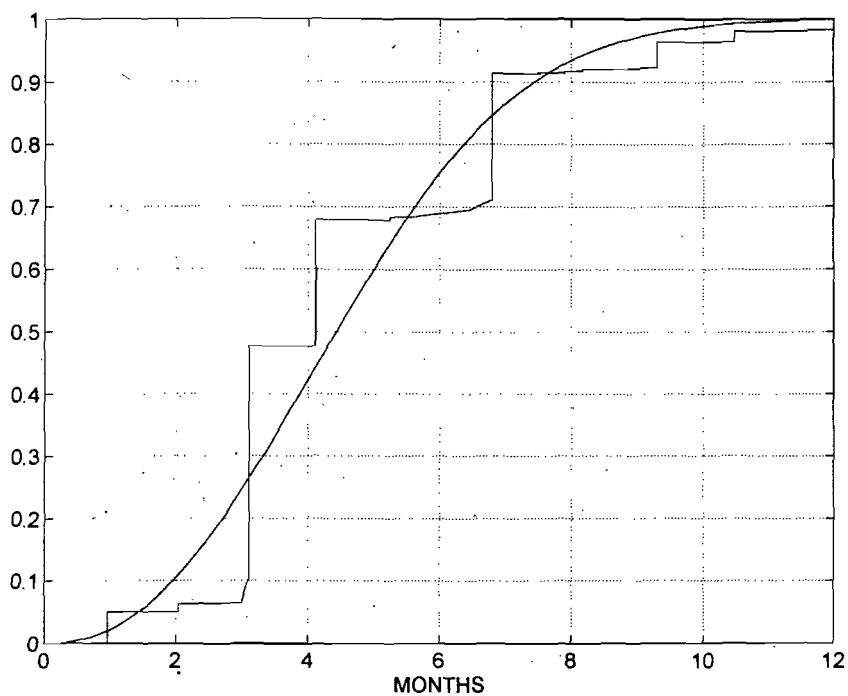


Figure 1. Duration of breastfeeding: Empirical and fitted cdf's

Then it follows that

$$g(z) = \begin{cases} p_1 & \text{for } z = 0 \\ ke^{-\lambda z} & \text{for } 0 < z < 5 \\ p_2 & \text{for } z = 5 \end{cases} \quad (2.4)$$

with cdf

$$G(z) = \begin{cases} p_1 & \text{for } z = 0 \\ p_1 + \frac{k}{\lambda} (1 - e^{-\lambda z}) & \text{for } 0 < z < 5 \\ 1 & \text{for } z \geq 5 \end{cases} \quad (2.5)$$

where

$$k = \frac{\lambda(1 - p_1 - p_2)}{1 - e^{-5\lambda}}.$$

To estimate p_2 , we use the official five year mortality rate for Lesotho (Lesotho Bureau of Statistics (1994)), which is 13.5%. This means that $\hat{p}_2 = 0.865$. From the data an estimator for p_1 follows as $\hat{p}_1 = 0.0563$, since there are 211 cases where $Z = 0$ out of the 3748 observations uncensored in X . Assuming p_1 and p_2 known, the maximum likelihood estimator of λ follows from (2.4) and (2.5) as $\hat{\lambda} = 0.9395$.

2.3 The joint distribution of X and Z

The Farlie-Morgenstern family of bi-variate distributions (Rice (1995) p. 71) is defined by

$$\begin{aligned} H(x, z) &= P[X < x, Z < z] \\ &= F(x)G(z) \{1 + \alpha(1 - F(x))(1 - G(z))\} \end{aligned} \quad (2.6)$$

where $F(x)$ and $G(z)$ are the marginal cdf's and $\alpha(-1 \leq \alpha \leq 1)$ is a measure of the dependence between X and Z . It is clear that independence is obtained when $\alpha = 0$.

Constructing a Farlie-Morgenstern type joint distribution from (2.3) and (2.5), it follows that

$$H(x, z) = \begin{cases} p_1 \left(1 - e^{-(x/\theta)^\beta}\right) \left\{1 + \alpha(1 - p_1)e^{-(x/\theta)^\beta}\right\}, & z = 0, 0 < x < \infty \\ \left[p_1 + \frac{k}{\lambda}(1 - e^{-\lambda z})\right] \left[1 - e^{-(x/\theta)^\beta}\right], & 0 < z < 5 \\ \left\{1 + \alpha e^{-(x/\theta)^\beta} \left(1 - p_1 - \frac{k}{\lambda}(1 - e^{-\lambda z})\right)\right\}, & 0 < x < \infty \\ 1 - e^{-(x/\theta)^\beta}, & z = 5, 0 < x < \infty \end{cases} \quad (2.7)$$

The joint density function is given by

$$h(x, z) = \begin{cases} p_1 f(x) \left\{1 + \alpha(1 - p_1)(2e^{-(x/\theta)^\beta} - 1)\right\}, & z = 0, 0 < x < \infty \\ k e^{\lambda z} f(x) \left\{1 + \alpha \left[1 - 2p_1 - 2\frac{k}{\lambda}(1 - e^{-\lambda z})\right] (2e^{-(x/\theta)^\beta} - 1)\right\}, & 0 < z < 5 \\ p_2 f(x) \left\{1 - \alpha(1 - p_2)(2e^{-(x/\theta)^\beta} - 1)\right\}, & z = 5, 0 < x < \infty \end{cases} \quad (2.8)$$

where $f(x)$ is given in (2.2).

Let

$$a(x) = 2e^{-(x/\theta)^\beta} - 1, \quad (2.9)$$

then the conditional distribution of Z given $X = x$, is given by

$$\begin{aligned} P[Z = 0|x] &= p_1[1 + \alpha(1 - p_1)a(x)] \\ g(z|x) &= ke^{-\lambda z} [1 + \alpha a(x) \{1 - 2p_1 - 2\frac{k}{\lambda}(1 - e^{-\lambda z})\}], \quad 0 < z < 5 \\ P[Z = 5|x] &= p_2[1 - \alpha(1 - p_2)a(x)]. \end{aligned} \quad (2.10)$$

The conditional expectation of Z is

$$E[Z|x] = k [C_1 + \alpha a(x) \{ (1 - 2p_1 - \frac{2k}{\lambda}) C_1 + \frac{2k}{\lambda} C_2 \}] + 5P[Z = 5|x], \quad (2.11)$$

where

$$C_j = \frac{1}{(j\lambda)^2} (1 - e^{-5j\lambda} - 5j\lambda e^{-5j\lambda}), \quad j = 1, 2.$$

The conditional survival function of Z is given by

$$\begin{aligned} S(z|x) &= P[Z \geq z|x] \\ &= P[Z = 5|x] + \frac{k}{\lambda} (e^{-\lambda z} - e^{-5\lambda}) [1 + \alpha a(x) \{ (1 - 2p_1 - \frac{2k}{\lambda}) + \frac{k}{\lambda} (e^{-\lambda z} + e^{-5\lambda}) \}] \\ &\quad \text{for } 0 < z < 5. \end{aligned} \quad (2.12)$$

3. Bayesian estimation of the parameters

In this section the Bayes estimators used in the analysis, are derived. Let

$$\pi(p_1, \theta, \beta, \lambda, \alpha) \propto \frac{1}{\lambda\beta}, \quad (3.1)$$

be the joint non-informative prior distribution on the unknown parameters where $\theta, \beta, \lambda > 0$, $0 < p_1 < 1$ and $-1 \leq \alpha \leq 1$. This means locally uniform priors on all parameters except λ and β , which have Jeffreys' priors, independent of the other parameters.

Let

$$\delta_i^z = \begin{cases} 1 & \text{if the } i\text{-th child has died} \\ 0 & \text{if the } i\text{-th child is still alive.} \end{cases}$$

Similarly for the indicator δ_i^x as defined in paragraph 2.1.

$$\text{Also, let } \delta_i^0 = \begin{cases} 1 & \text{if } z_i = 0 \\ 0 & \text{if } z_i > 0. \end{cases}$$

The likelihood function is then given by

$$L(p_1, \theta, \beta, \lambda, \alpha | \underline{x}, \underline{z}, \underline{\delta}^x, \underline{\delta}^z, \underline{\delta}^0)$$

$$\alpha \prod_{i=1}^n [g(z_i | x_i) f(x_i)]^{\delta_i^x \delta_i^z (1 - \delta_i^0)} [P[Z_i = 0 | x_i] f(x_i)]^{\delta_i^x \delta_i^z \delta_i^0} \times \\ [S(z_i | x_i) f(x_i)]^{\delta_i^x (1 - \delta_i^z)} [P(X_i \geq x_i, Z_i \geq 0)]^{(1 - \delta_i^x)(1 - \delta_i^z)}. \quad (3.2)$$

The four terms correspond to the cases where neither variable is censored with $z_i > 0$ and with $z_i = 0$, where z only is censored and where both variables are censored. The functions are defined in (2.2), (2.10) and (2.12). The number of observations corresponding to the four terms in the likelihood function are, respectively, 182, 211, 3355 and 253.

The joint posterior distribution is now proportional to the product of equations (3.1) and (3.2). By means of the Gibbs sampler (Gelman, et. al. (1995)) and the acceptance-rejection method, samples of 10 000 were drawn from each of the marginal posterior distributions of the five parameters. In

table 1 the posterior means and 95% highest posterior density (HPD) intervals of the parameters are given.

Table 1. Posterior means and 95% HPD intervals

Parameter	Post. mean	95% HPD interval
p_1	0.0737	0.0448 – 0.1014
θ	0.4164	0.3797 – 0.4559
β	2.1804	1.9781 – 2.4108
λ	1.0413	0.5603 – 1.7923
α	0.8720	0.6310 – 1.0000

It is of interest to compare the estimators in table 1 with the maximum likelihood estimators obtained from the marginal likelihoods in section 2. In figure 2 the marginal posterior distribution of α is shown. This indicates a strong positive relationship between X (duration of breastfeeding) and Z (survival time).

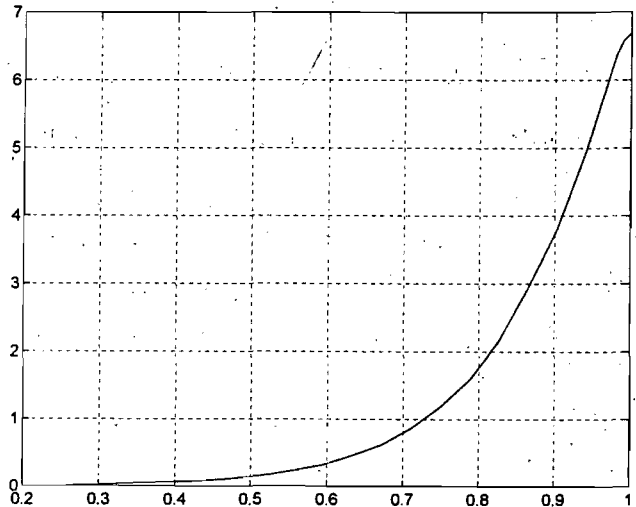


Figure 2. Marginal posterior density function of α

4. Survival curves

From equation (2.10) with the parameters as in table 1, the probabilities $P[Z = 0|x]$ (child dies during period of breastfeeding) and $P[Z = 5|x]$ (child survives beyond the age of five) are shown in figure 3 as a function of x , the duration of breastfeeding.

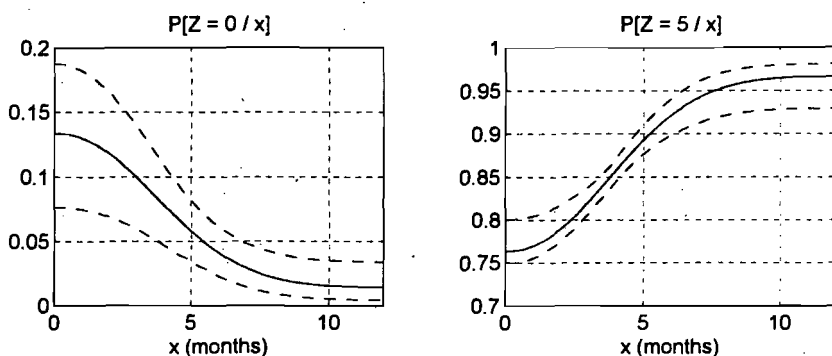


Figure 3. $P[Z = 0|x]$ and $P[Z = 5|x]$ with 95% HPD intervals

From paragraph 2.2 the unconditional probability $P[Z = 5]$ is assumed known as $p_2 = 0.865$, while the Bayes estimate of $P[Z = 0]$ is $p_1 = 0.0737$. The conditional probabilities in figure 3 show the effect of x on these probabilities. Consider again the actual lifetime $Y = Z + X$. A linear transformation on (2.12) yields the survival function of Y conditional on x . In

figure 4 the survival curves $P[Y \geq y|x]$ are shown for $x = 1, 4$, and 9 months of breastfeeding.

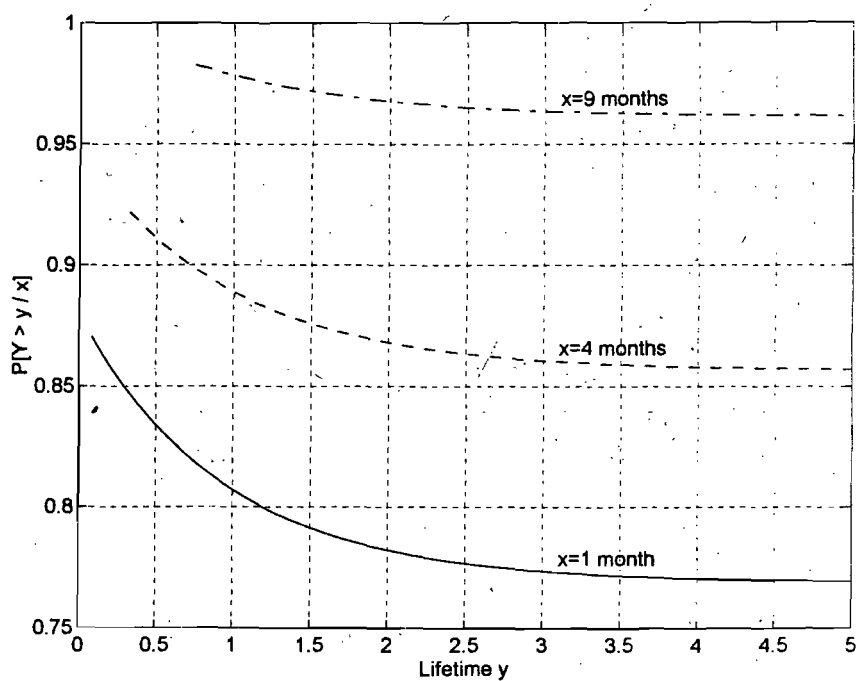


Figure 4. Life survival curves for $x = 1, 4$ and 9 months

From the figure it can be seen that there is a substantial difference in the probabilities of survival between a child with 1 month of breastfeeding and one with 9 months of breastfeeding. For $y = 5$ years this difference is $0.961 - 0.769 = 0.192$. A large proportion of this difference is due to the fact that $P[Z = 0|x]$ is a decreasing function of x , as seen in figure 3. A child with 1 month of breastfeeding has less chance of surviving the termination of breastfeeding than a child with 9 months of breastfeeding (0.870 against 0.982). To determine the effect of length of breastfeeding on the continued survival of a child we should also compare survival curves conditional on the child surviving the termination of breastfeeding.

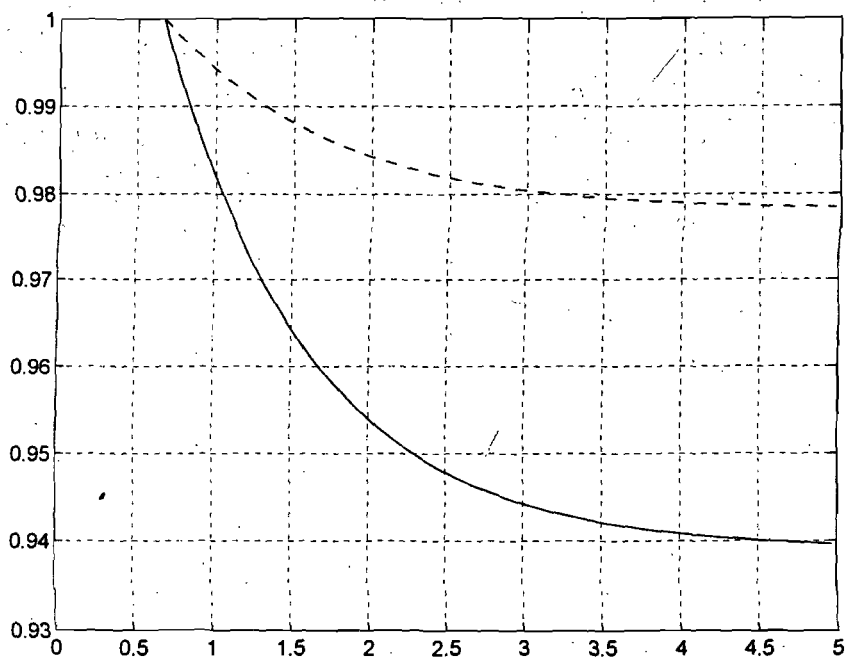
So for $x_1 > x_2$, we want to compare

$$P[Z \geq z | X = x_1, Z > 0] = \frac{S(z|x_1)}{1 - P[Z = 0|x_1]} \quad (3.1)$$

with

$$P[Z \geq z | X = x_2, Z \geq x_1 - x_2] = \frac{S(z|x_2)}{S(x_1 - x_2|x_2)}, \quad (3.2)$$

the survival curve of a child with breastfeeding period of length x_2 , but with survival at least up to x_1 . In figure 5 the survival curve of a child with 1 month of breastfeeding and a survival time of 9 months is compared with that of a child surviving beyond 9 months of breastfeeding. The figure is scaled in terms of the actual lifetime Y .



Lifetime y (years)

Figure 5. Survival curves $P[Y \geq y | X = 9 \text{ months}, Y > X]$ (---) and $P[Y \geq y | X = 1 \text{ month}, Y > 9 \text{ months}]$ (—)

The survival rate for $X = 9$ months is still better, but the difference at $y = 5$ is now $0.979 - 0.940 = 0.039$, showing a relatively small effect of duration of breastfeeding on survival rate once the termination of breastfeeding is overcome.

5. Concluding remarks

The approach that we followed to determine the relationship between breastfeeding duration and lifetime of children is to start with the marginals and then construct the joint distribution. To eliminate the built-in dependence between X and Y because of the restriction that $X \leq Y$, we consider the variables X and $Z = Y - X$. For this model with Bayes estimators, the correlation coefficient between X and Z is $\rho_{XZ} = 0.1687$, and, after transformation, the correlation between X and Y is $\rho_{XY} = 0.2752$. The difference can be attributed to the condition that $X \leq Y$, and $\rho_{XY} > \rho_{XZ}$ always. With the other parameters fixed, the possible range for ρ_{XY} is $-0.113 \leq \rho_{XY} \leq 0.299$ as α moves from -1 to $+1$.

The results in section 3 show strong evidence that $\alpha > 0$ and thus a positive relationship between X and Z . We must, however, be careful in our causal interpretation of these results. When the period of breastfeeding is short, Z is more likely to be zero as seen in figure 3. However, in the large majority of these cases it is the death of the child that curtailed the period of breastfeeding and not the termination of breastfeeding that contributed to the death of the child. That is why the conditional survival curves, as shown in figure 5, give a more realistic picture of the comparative survival rates. Our conclusion is thus

that there is a significant positive relationship between duration of breastfeeding and the survival rate of children.

We believe that the Farlie-Morgenstern family of joint distributions provides a useful method for constructing a model for analyzing two variables jointly and can make a contribution to further demographical studies in this area.

6. References

- GELMAN, A., CARLIN, J.B., STERN, H.S. & RUBIN, D.B. (1995). *Bayesian Data Analysis*. Chapman-Hall, London.
- GUMMER-STRAWN, L.M. (1993). Regression Analysis of Current-Status Data: An Application to Breast-feeding. *J. Am. Statist. Ass.*, **88**, 758–765.
- KAPLAN, E.L. & MEIER, P. (1958). Nonparametric estimation from incomplete observations. *J. Amer. Statist. Assoc.*, **53**, 475–481.
- LESOTHO BUREAU OF STATISTICS (1994): Lesotho Statistical Year Book. Bureau of Statistics, Maseru, Lesotho.
- MCDÓNALD, J. (1981). *A new methodological approach for the analysis of WFS current-status breast-feeding data*. WFS/Tech, 1732, London: World Fertility Surveys.
- RICE, J.A. (1995). *Mathematical statistics and data analysis*, 2nd ed. Duxbury Press.
- VANDERHOEFT, C. (1982). Accelerated failure time models: An application to current-status breast-feeding data from Pakistan. *Genus*, **38**, 135–157.
- WORKU, Z.B. (1995). *Special report on: Factors that influence child survival at the Maseru districts*. Department of Statistics and Demography, The National University of Lesotho.